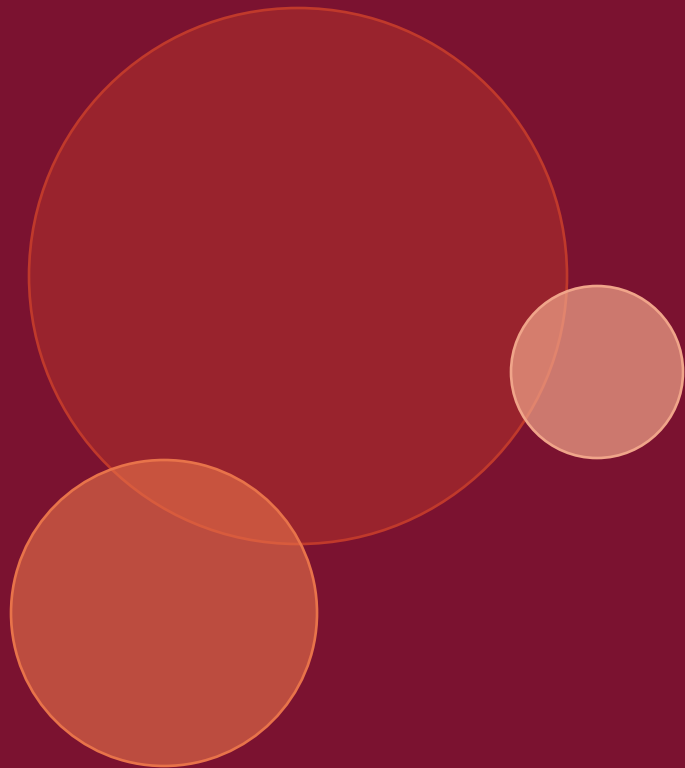


USDA Pesticide Analysis

Lauren Myers | [GitHub Repository](#)



Introduction

Intro

Data sourced from USDA PDP

Years Span From 1994-2024

164,112 Tested Samples

365 Unique Pesticides

Key Findings

2.94 Pesticides Per Sample (All Years)

365 Unique Pesticides Found (All Years)

Top Ten Pesticides Found (2018-2024)

Fludioxonil

Azoxystrobin

Pyraclostrobin

Boscalid

Fluopyram

Pyrimethanil

Thiabendazole

Imidacloprid

Acetamiprid

Methoxyfenozide

Top Ten Pesticides

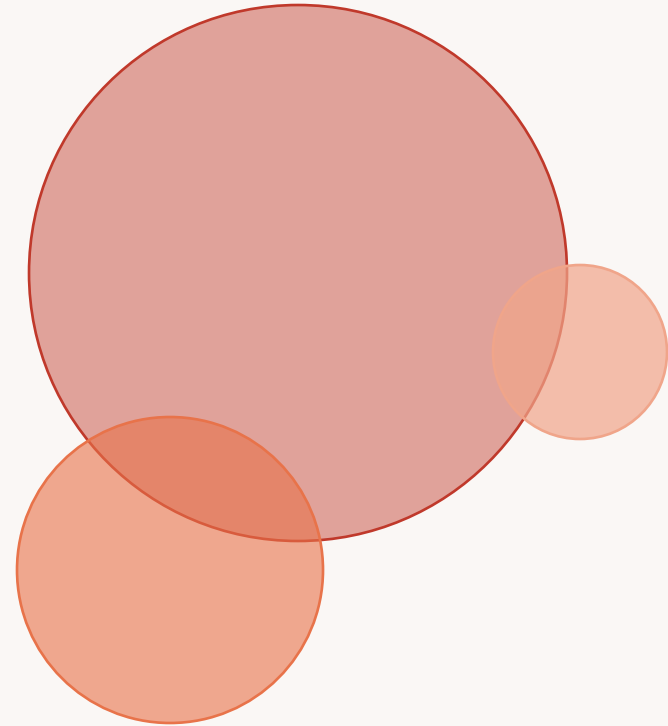
Broad-Spectrum Fungicides

6 of the 10 are this category : Fludioxonil, Azoxystrobin, Pyraclostrobin, Boscalid, Fluopyram, and Pyrimethanil

All Metabolize Through liver detoxification

2024 European Food Safety Authority Classified Fludioxonil as Endocrine Disruptor (EFSA, 2024)

Concern drawn from mixture-based pesticide risk

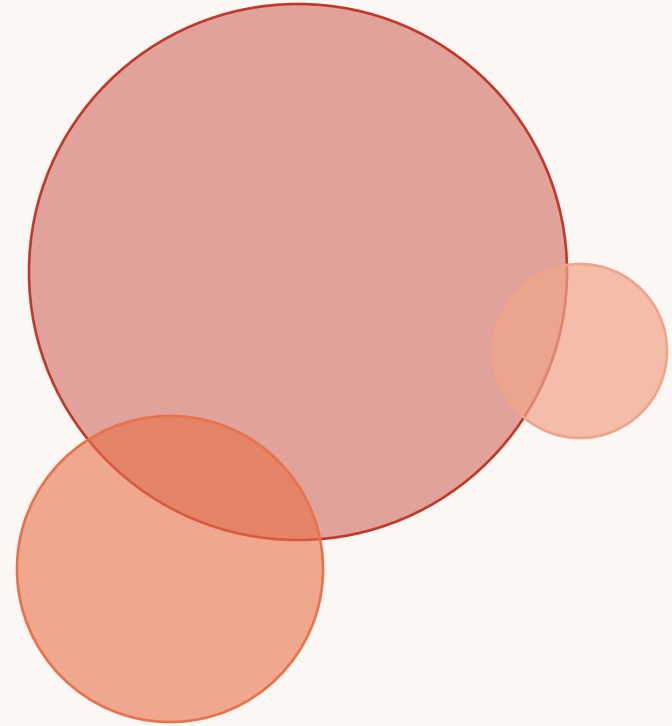


Thiabendazole

Banned in EU as a Food Additive (EFSA, 2022)

Determined Likely Carcinogenic in high doses

Endocrine disruption through thyroid



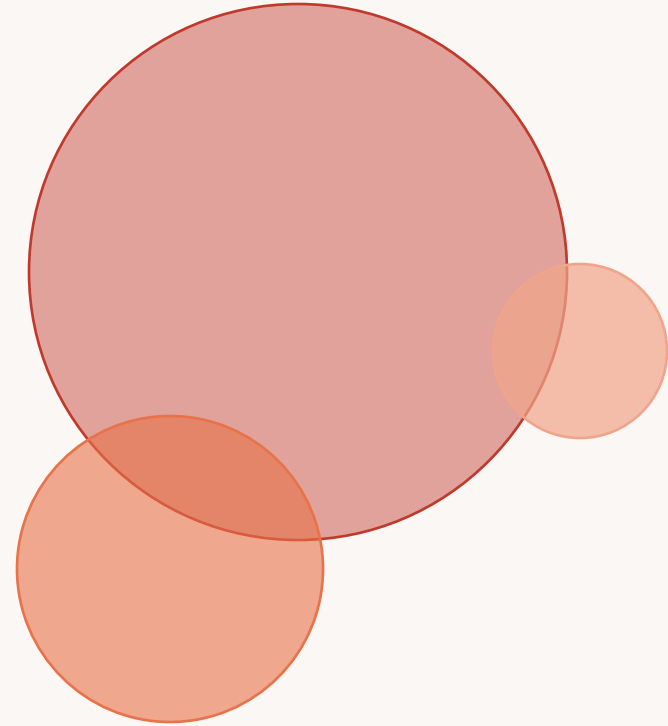
Neonicotinoids: Imidacloprid and Acetamiprid

Concerns Raised to Toxic Nature to Bee Populations
(Ahsan, Z et al. 2025)

Study of Rats show high doses cause reduction in
Brain volume (Sass et al. 2024)

Cited especially effects to the developing brain

EPA flagged that risks were ignored



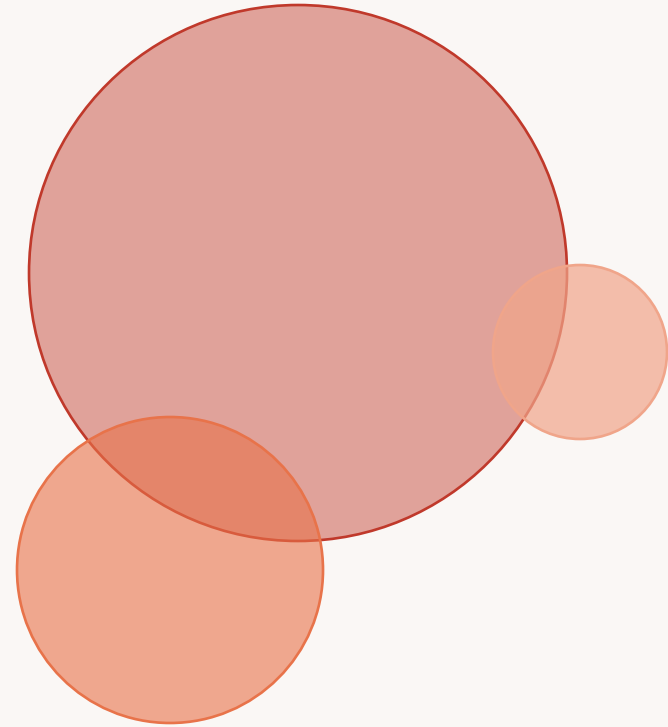
Methoxyfenozide

Highly selective to insects

Insect growth regulator

Mimics a molting hormone in insects

Not a high risk to humans



Pesticide Problems

Issues Beyond Residue In Produce

Pesticide Mixtures

- 2.94 Pesticides Per Sample

- Raises Concern with Compound Effects

Pesticides In Water

- Study of 1200 wells tested

- 40% found pesticide residues

- Two-thirds had mixtures

Pesticide Resistance

- Pests becoming more resistant to pesticides

- Requiring new pesticides to be made

Potential Solutions

Organic Consumption

Studies Found Reduction in Pesticide Residue in humans

In as quickly as five days

Remained undetected until conventional diet was reintroduced

Other study found an overall pesticide residue reduction of up to thirty percent

Regenerative Farming and Non-toxic Alternatives

Reduces the sustained pesticides in the soil by breaking them down quicker

A possibility for farmers that prevents the use Of synthetic pesticides for crops

Microbial Pesticides offer an alternative solution to the broad-spectrum synthetic pesticides

Wrap Up

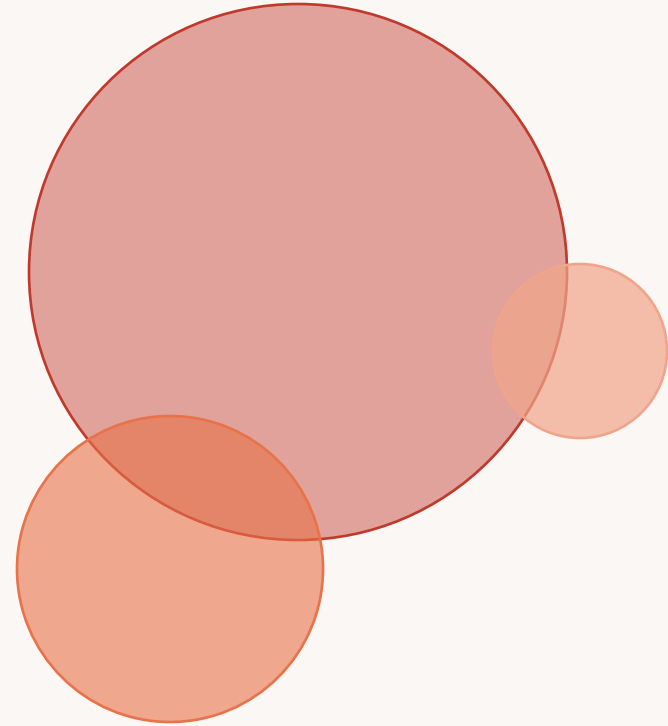
Wrap Up

Global Pesticide Use has increase by 50% in the last three decades (Hamed et al., 2025)

Growing concerns of effects on synthetic pesticides and their long-term effects on humans remain somewhat unknown

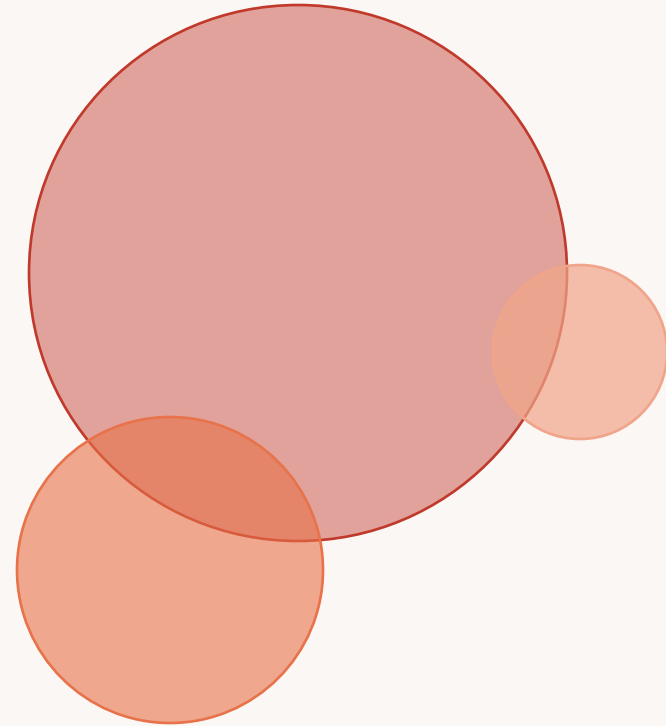
Organic and Regenerative Farming are promising solutions, but continued research is needed to study the effects of pesticides in humans and the environment

[GitHub Repository](#)



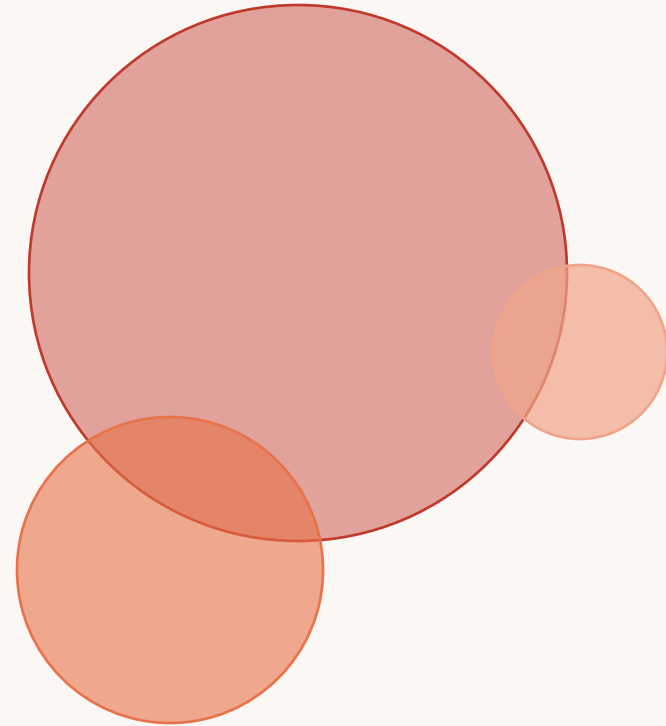
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